



CS & IT ENGINEERING



Algorithms

Analysis of Algorithms

Lecture No.- 10



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

TC analysis



Topics to be Covered



Topic

Topic

Topic

TC analysis Contd...



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe



Topic : Analysis of Algorithms

#Q. Algo AJ(n)

```
{  
    if (n==2)  
        return 2  
    return (AJ( $\sqrt{n}$ ) + AJ( $\sqrt{n}$ ))  
}
```

Handwritten annotations:
- A green arrow points from $AJ(n)$ to $T(n)$.
- A green bracket groups the base case, with a 'b' next to it.
- Green arrows point from $AJ(\sqrt{n})$ to $T(\sqrt{n})$ for both occurrences in the recursive call.

54%

- A** $O(n)$
- B** $O(\sqrt{n})$
- ☒ **C** $O(\log n)$
- D** $O(\log(\log n))$

Step 1:-

$$T(n) = T(\sqrt{n}) + T(\sqrt{n}) + a, n > 2$$

$$T(n) = 2T(\sqrt{n}) + a, n > 2$$

$$T(n) = b, n = 2$$

Step 1:- $T(n) = 2T(\sqrt{n}) + a$ — (1)

$$T(n) = 2T(n^{1/2}) + a \quad \rightarrow (2^1 - 1)a$$

$$T(n^{1/2}) = 2T(n^{1/2^2}) + a$$

$$T(n) = 2(2T(n^{1/2^2}) + a) + a$$

$$T(n) = 2^2 T(n^{1/2^2}) + 2a + a$$

$$T(n) = 2^2 T(n^{1/2^2}) + 3a$$

$$T(n^{1/2^2}) = 2T(n^{1/2^3}) + a$$

$$T(n) = 2^2 T(n^{1/2^2}) + 3a \quad \rightarrow (2^2 - 1)a$$

$$T(n) = 2^3 T(n^{1/2^3}) + 7a$$

$$\downarrow$$

$$(2^3 - 1)a$$

generalised

$$T(n) = 2^k T(n^{1/2^k}) + (2^k - 1) a$$

For Base Condition

$$n^{1/2^k} = 2$$

$$\frac{1}{2^k} \log n = 1$$

$$\boxed{2^k = \log n}$$

$$T(n) = \log n * T(2) + (\log n - 1) a$$

$$\boxed{T(n) = b * \log n + a * \log n - a}$$

Step 3: $T(n) = O(\log_2 n)$



```
#Q. Algo AJ(n) AJ1(n)
      {
        if (n==1)
          return 2
        else
          return (AJ 1(n/2) + AJ1(n/2) + AJ2(n))
      }
}
```

$\hookrightarrow T(n)$

$\downarrow$ $\downarrow$ $\Downarrow$
 $T(n/2)$ $T(n/2)$ $O(1)$

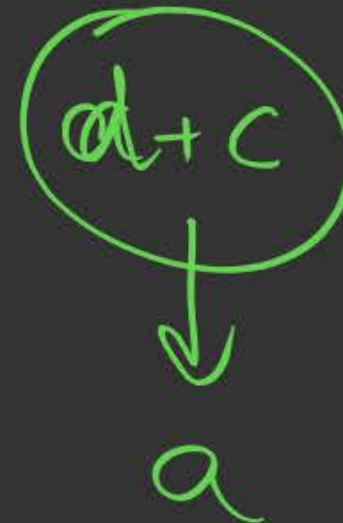
85%
↓
80%

- A** $O(\sqrt{n})$
- B** $O(n)$
- C** $O(n^2)$
- D** $O(\log n)$

Step 1:-

$$T(n) = 2T(n/2) + a, \quad n > 1$$

$$T(n) = b, \quad \underline{n = 1}$$



Step 2:-

$$T(n) = 2T(n/2) + a \text{ --- (1)}$$

$$T(n/2) = 2T(n/2^2) + a$$

$$T(n) = 2^2 T(n/2^2) + 3a \text{ --- (2)}$$

$\rightarrow (2^2 - 1)a$

$$T(n) = 2^3 T(n/2^3) + 7a \text{ --- (3)}$$

$$\hookrightarrow (2^3 - 1)a$$

In genal,

$$T(n) = 2^k T(n/2^k) + (2^k - 1)a$$

For B.C,

$$\frac{n}{2^k} = 1$$

$$\Rightarrow 2^k = n$$

$$\Rightarrow T(n) = nT(1) + (n-1)a$$

$$T(n) = n \times b + (n-1)a \rightarrow \text{value}$$

$$\begin{aligned} 2^k &= 2^{\log_2 n} \\ &= n^{\log_2 2} \\ &= n \end{aligned}$$

Step 3 :-

$$T(n) = O(n)$$



Topic : Analysis of Algorithms

#Q. Algo AJ(n)

{

if (n == 1)

return 1

else

{

AJ (n/2)

AJ (n/2)

AJ 2 (n) $\Rightarrow$ $O(n)$

}

}

V. Imp

$\Rightarrow$ Merge Sort

90%

↓
45%

A

$O(n)$

B

$O(n \log n)$

C

$O(\log n)$

D

$O(n^2 \log n)$

Step 1:

$$T(n) = T(n/2) + T(n/2) + n + a$$

$$T(n) = 2T(n/2) + n, n > 1$$

$$T(n) = \underline{b}, n = 1$$

Step 1: $T(n) = 2T(n/2) + n$ — (1)

$$T(n/2) = 2T(n/2^2) + n/2$$

$$T(n) = 2(2T(n/2^2) + n/2) + n$$

$$T(n) = 2^2 T(n/2^2) + 2n$$
 — (2)

$$T(n) = 2^3 T(n/2^3) + 3n$$
 — (3)

general term

$$T(n) = 2^k T(n/2^k) + k \times n \quad \text{--- (4)}$$

For B.C, $\frac{n}{2^k} = 1$

$$2^k = n$$

$$k = \log_2 n$$

$$T(n) = n \times T(1) + n \times \log_2 n$$

$$T(n) = n \times b + n \times \log_2 n \rightarrow \text{value.}$$

Step 3:-

$$T(n) = O(n \log_2 n)$$



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V. adv

#Q. Given a recurrence, determine the time complexity.

$$T(n) = 2, n = 2$$

$$T(n) = \sqrt{n} * T(\sqrt{n}) + n, n > 2$$

75%
↓
68%

A $O(n \log n)$ ✗

B $O(n \log(\log n))$ ✓

C $O(n^2)$ ✗

D $O(\log(\log n))$ ✗

Ⓟ

$$T(n) = n^{(\frac{1}{2} + \frac{1}{2^2})} \underbrace{T(n^{\frac{1}{2^2}})}_{\text{from (1)}} + 2n \quad \text{--- (2)}$$

$$T(n^{\frac{1}{2^2}}) = n^{\frac{1}{2^3}} T(n^{\frac{1}{2^3}}) + n^{\frac{1}{2^2}}$$

$$T(n) = n^{(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3})} T(n^{\frac{1}{2^3}}) + 3n \quad \text{--- (3)}$$

★

genral: $T(n) = n^{\left(\frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^k}\right)} T(n^{1/2^k}) + k * n$

$$\frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^k}$$

$\Rightarrow$ GP: $a = 1/2$
 $r = 1/2$
 $n = k$

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$S_n = \frac{\frac{1}{2} \left(1 - \frac{1}{2^k}\right)}{1 - 1/2}$$

$$S_n = \left(1 - \frac{1}{2^k}\right)$$

$$T(n) = n^{\left(1 - \frac{1}{2^k}\right)} T(n^{1/2^k}) + k * n \quad \text{--- (4)}$$

For B.C,

$$n^{1/2^k} = 2$$

$$\frac{1}{2^k} \log n = 1$$

$$\log n = 2^k \Rightarrow k = \log(\log n)$$

↓

$$T(n) = n^{(1 - 1/2^k)} T(n^{1/2^k}) + k * n$$

$$= \frac{n}{n^{1/2^k}} T(n^{1/2^k}) + k * n$$

$$\underline{\underline{n^{1/2^k} = 2}}$$

$$= \frac{n}{2} * T(2) + n * \log(\log n)$$

$$= \frac{n}{2} * 2 + n \log(\log n)$$

$$\boxed{T(n) = n + n \log(\log n)} \rightarrow \underline{\text{value}}$$

$$\underline{\text{TC: } O(n \log(\log n))}$$

Doubt

$$n^2$$

$$n \log \log n$$

$$n \times n$$

$$n \times \log \log n$$

$$n >$$

$$\log \log n$$



Topic : Analysis of Algorithms

Algo AJ(n)

V. Imp?

{

if (n == 2)

return 1

else

return [AJ($\sqrt{n}$) + 10]

}

$$T(n) = T(\sqrt{n}) + a, n > 2$$

$$T(n) = b, n = 2$$

$$T(n) = T(n^{1/2}) + a \quad \text{--- (1)}$$

$$T(n^{1/2}) = T(n^{1/2^2}) + a$$

$$T(n) = T(n^{1/2^2}) + 2a \quad \text{--- (2)}$$

$$T(n) = T(n^{1/2^3}) + 3a \quad \text{--- (3)}$$

$$T(n) = T(n^{1/2^k}) + k a$$

$$n^{1/2^k} = 2$$

$$2^k = \log n$$

$$k = \log(\log n)$$

$$\rightarrow T(n) = T(1) + \log(\log n) a$$

$$T(n) = b + \log(\log n) a$$

$$\underline{T(n) = O(\log(\log n))}$$



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Algo AJ(n)

{

 if (n == 1)

 return n

 else

 return (AJ (n/2) + 10)

}

V. Imp

$\Rightarrow$

Binary Search

$$T(n) = b, n = 1$$

$$T(n) = T(n/2) + a, n > 1$$

$$T(n) = T(n/2) + a \quad \text{--- (1)}$$

$$T(n/2) = T(n/2^2) + a$$

$$T(n) = T(n/2^2) + 2a \quad \text{--- (2)}$$

$$T(n) = T(n/2^3) + 3a \quad \text{--- (3)}$$

$$T(n) = T(n/2^k) + k \times a \quad \text{--- (4)}$$

$$T(n) = T(1) + a \times \log n$$

$$T(n) = b + a \times \log n$$

for B.C, $\frac{n}{2^k} = 1 \Rightarrow 2^k = n$

$$k = \log_2 n$$

$$T(n) = O(\log_2 n)$$



Summary



TC analysis of Recursive Algo



THANK - YOU